A Pilot That Learns and a Twin That Watches: Testing a Cognitive Pilot Model Against Pilots It Has Never Seen

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Abstract. Flight instructors judge a trainee from the seat behind, one flight at a time. A digital twin of the pilot could follow every flight against the written procedure and anticipate the trainee's next actions, provided it can do so for a pilot it has never seen. We test this on VC-PILOT, an ACT-R pilot whose procedures are held as declarative knowledge drawn from a procedure ontology. Read in reverse, the same knowledge becomes a mirror that recognises which step of the procedure a pilot is executing; run twenty seconds ahead of the aircraft, it becomes a twin that predicts when the pilot's next actions will come. On 211 simulator take-offs by twelve pilots, nominal and with engine failures, the mirror recognised all 1,686 pilot actions found offline, with a median lag of 0.08 s. Predicting each pilot from that pilot's earlier flights alone, with everything else learned from the other eleven, the twin came closer to the pilot's actual timing than the pilot's own record did for 11 of 12 pilots, by 1.0 s at a 20 s horizon, and closer than the strongest baseline that follows the flight without a cognitive model for 9 of 12, by 0.32 s (95 % interval 0.12 to 0.57), an advantage carried by the cognitive model itself. At 10 s it is not established. The pacing of routine actions proved to be a personal signature, with intraclass correlations of 0.71 to 0.80, while the response to a failure was not, which bounds what an expectation-based monitor can flag. Each pre-registered criterion we discuss is reported with its verdict, failures included.

Keywords: Cognitive digital twin · ACT-R · Pilot modelling · Individual differences · Pre-registration · Flight simulation

1 Introduction

In flight training the instructor is the measuring instrument. Seated behind the trainee, the instructor follows the procedure step by step, notices that the gear came up late or that the flaps were retracted early, and anticipates the next mistake before it happens. Simulators record every flight, but a recording does not say which step of the procedure the trainee was on, whether an action came late, or what the trainee was about to do. Those are the questions an instructor

answers, and they are what a digital twin of the pilot would have to answer to be of use to a training organisation.

Such a twin has to meet three demands. It must *follow* a pilot's flight closely enough to name the step being executed. It must *anticipate* the timing of the pilot's next actions, since a missing action can only be flagged once it is overdue. And it must do both for a pilot it has *never seen*, because a trainee is new by definition. A fourth demand concerns the model behind the twin rather than the twin itself: if the same model is to stand in for trainees when a procedure or a cockpit is redesigned, its skill should improve with practice the way a person's does.

Cognitive architectures have flown aircraft in simulation for a long time [9, 22], and some have been compared with pilots [21]. The comparison is usually made on group averages, and the few models that follow individual pilots trace their behaviour in specific situations, such as their response to an alert [11, 2]. A twin is judged differently: on one person at a time, over whole procedures, and against a demanding alternative. Whatever a cognitive model predicts about a pilot, one could also predict from that pilot's own past flights, or from the events of the current flight, without any cognitive model at all. A twin earns its keep only if it does better than these.

We build on VC-PILOT [10], an ACT-R pilot [1] whose procedures are not written as production rules but held as declarative knowledge, read from a procedure ontology [4] by a small, fixed set of generic rules. Because the procedure is data, the same knowledge can be used three ways: to fly the aircraft, to read a pilot's flight against the procedure (the *mirror*), and to run ahead of the aircraft and say what the pilot will do next (the *twin*). We ask four questions. Does the skill the model acquires through practice match pilots it was never tuned on? Can the mirror follow a pilot's flight? Does the twin anticipate a pilot better than the pilot's own record, and better than a strong statistical baseline that knows what has already happened in the flight? And what, in an individual pilot's timing, is predictable at all?

The paper makes four contributions. First, a mirror that reads a pilot's flight through the knowledge the agent flies with, so that a deviation is reported against a named step of the procedure rather than as an anomaly score. Second, a pilot-by-pilot evaluation of the twin against the pilot's own record and against the strongest baseline we could build without a cognitive model, in which each pilot is predicted from earlier flights only and everything else is learned from the other pilots; looking twenty seconds ahead, the advantage belongs to the cognitive model itself. Third, the observation that the pacing of routine actions is a stable personal signature while the response to a failure is not, and what this implies for training and for monitoring. Fourth, a record of pre-registered criteria with their failures: the model's learning plateaued against held-out pilots, and alerts on omitted actions fell short of their target. Every criterion was written, with its threshold, and time-stamped in version control before it was computed [14]; amendments are dated, and a failure is never removed by a change made to pass it.

2 Related Work

Cognitive pilots. TacAir-Soar, a rule-based system built in Soar [13], flew most of the fixed-wing missions of the United States military in large simulated exercises [9]. In ACT-R [1, 15], a model of commercial jetliner taxiing flew among aircraft replayed from recordings of a real airport, so that its trajectories could be compared with those of pilots [22]; QN-ACTR extended the architecture toward multitask performance [3]. Xu and colleagues compared two models of pre-flight and take-off with pilots flying the same tasks in a simulator, with errors of 0.8 to 13.2 % on performance and workload [21]. Kotseruba and Tsotsos review what such architectures can and cannot yet do [12]. Our earlier work moved a pilot's procedures out of production rules into a procedure ontology, so that one interpreter flies different aircraft [10]. Closest to our question, Klaproth and colleagues traced individual pilots' responses to flight-deck alerts with an ACT-R model informed by EEG [11], and Blum and colleagues anticipated the actions of thirteen pilots with an ACT-R model that represents individual differences [2]. To our knowledge, no such model has been compared, pilot by pilot and with each pilot held out, against the pilot's own record and against a baseline that follows the flight, as a predictor of when the next actions will come. That is the test we apply here.

Skill acquisition. Production compilation turns the slow interpretation of declarative instructions into fast task-specific rules [19]; threaded cognition explains how concurrent tasks share the same resources [17]. Both bear on whether a model can improve with practice the way a trainee does, the question of our first test.

Inferring the pilot's state. Neuroadaptive and assistance systems infer the pilot's mental state from physiology and behaviour [11, 16], in the line of work on situation awareness [6] and workload [8, 20]. ACT-R models have represented the effect of fatigue on performance [7]. We test whether an estimated state improves the twin's predictions; it does not (Section 6).

Individuals and populations. Predicting an individual from few observations is a classic problem of empirical Bayes: an individual estimate is shrunk toward the population in proportion to how much individuals actually differ [5]. The same variance decomposition gives the intraclass correlation [18], which we use to ask which parts of a pilot's timing are personal.

3 One Body of Knowledge, Three Uses

3.1 Procedures as declarative knowledge

A procedure is a set of tasks. Each task carries one or more *actions*, a cockpit parameter and the value it must reach (thrust levers at take-off thrust, gear lever up), *preconditions* on earlier tasks, and *constraints* on the aircraft's state (a speed, a height). Tasks belong to a procedure: the normal take-off, the rejected

Table 1. The aviation terms used in this paper.

Term	Meaning
Brake release	Start of the take-off roll; time zero of a take-off in this paper.
Take-off thrust	Thrust levers set to take-off power, after a stabilisation at about 50 %.
V1	Decision speed: below it a failure leads to a rejected take-off, above it the take-off continues.
Rotation, liftoff	The pilot raises the nose; the wheels leave the ground.
Gear up, thrust reduction	Landing gear retracted after liftoff; thrust reduced to climb power at a set height.
Flaps 1, flaps 0	Successive retraction of the wing flaps as speed increases.
Rejected take-off	Take-off abandoned on the runway: thrust to idle, reverse thrust, maximum braking.
Continued take-off	Take-off pursued on one engine after a failure above V1.

take-off, and the take-off continued after an engine failure past the decision speed. Table 1 gives the few terms of aviation the paper needs.

A fixed set of 37 generic production rules interprets these tasks [10]. They retrieve the next task whose preconditions hold, wait for its constraints, act, and verify that the action took effect; no rule names a task. Practice changes this behaviour through two standard ACT-R mechanisms. Production compilation fuses the retrieval of a task and its execution into a task-specific rule nobody wrote [19], and utility learning, driven by a reward at the end of each task, decides when the compiled rule wins over the generic rules it came from. When a failure is detected, the interpreter switches procedure, for instance from the normal to the rejected take-off.

3.2 The mirror: reading a flight against the procedure

The mirror runs the same knowledge without acting. It reads the aircraft's state, from a recording or live from the simulator, and writes nothing back. For the task the pilot should be on, it compares each action's parameter with its target; an action counts as done once the parameter has reached its target and held it for one second, so that a hesitation shorter than a second does not break it. For every task the mirror logs when it was done and whether it came early, late or not at all with respect to qualified pilots, and when a failure appears it follows the pilot into the corresponding procedure. Offline, it reads recorded flights; live, an ACT-R model follows the flight in parallel, as the agent would fly it.

3.3 The twin: running the model ahead of the aircraft

The twin makes two kinds of prediction. The first is immediate: when the mirror's model reaches a task, it predicts the pilot's action at the time the model is ready,

plus an offset learned for that pilot. The offset is the pilot's median deviation on earlier flights, shrunk toward the median of other pilots as

$$J = \frac{n\,\tilde{x}_{\text{pilot}} + K\,\tilde{x}_{\text{others}}}{n + K}, \tag{1}$$

where $\tilde{x}$ denotes a median, n is the number of the pilot's earlier observations, and $K = 16$ was chosen on the calibration pilots alone. With few flights the pilot is predicted like everyone else; with many, from the pilot's own flights.

The second prediction looks ahead. Every five seconds a copy of the model, reset to the mirror's current position in the procedure, runs twenty seconds ahead in simulated time. It flies against an aircraft trajectory extrapolated from the five most similar take-offs of other pilots, and its commands go to a stand-in simulator, never to the real one. The time at which the copy is ready for each task, plus the pilot's offset for that task, is the twin's prediction. Two further details matter. The recorded flights are timed in simulated seconds, and the simulator in which the pilots flew ran about 9 % slower than the clock, whereas ACT-R's latencies are human seconds; the trajectory is therefore stretched by the ratio measured on each flight. And the prediction in service is a weighted average, $H = 0.75\,A + 0.25\,B$, of the model's prediction A and a reference B, the pilot's shrunk median delay of the action since brake release, or since the failure for the actions of a failure procedure; the weight was chosen on calibration pilots. Each prediction carries an interval meant to contain the action four times out of five, learned on the same pilots; on the validation pilots it contained 70 % of their actions, at the lower edge of the range we had required. Figure 1 shows what the twin said, ten seconds ahead, during one take-off of a pilot it had not been calibrated on.

The mirror and the twin run live next to the simulator, in a dashboard that shows the step the pilot is on, the predicted time of the next actions with their intervals, and the actions that are overdue. We have run it in live sessions with our own agent at the controls; it has not yet watched a human pilot live.

4 Data and Protocol

Pilots and flights. Twelve pilots flew take-offs in a fixed-base Airbus A320 simulator: five type-rated airline pilots, two licensed pilots and five novices. The corpus holds 211 take-offs, 104 nominal, 63 rejected after an engine failure and 44 continued on one engine; each pilot flew between 4 and 21 of them. The time of each failure is dated from the engines' fuel flows. A separate test of the learning model used 129 nominal take-offs from the same study. The flight data and the procedure ontology belong to the project's industrial partners and cannot be released.

Calibration, validation and blindness. Six pilots (2, 3, 5, 6, 7 and 10) served for calibration, and everything that can be tuned was tuned on them. The other six (4, 8, 9, 11, 12 and 13) were held out. They were judged blind for the last time

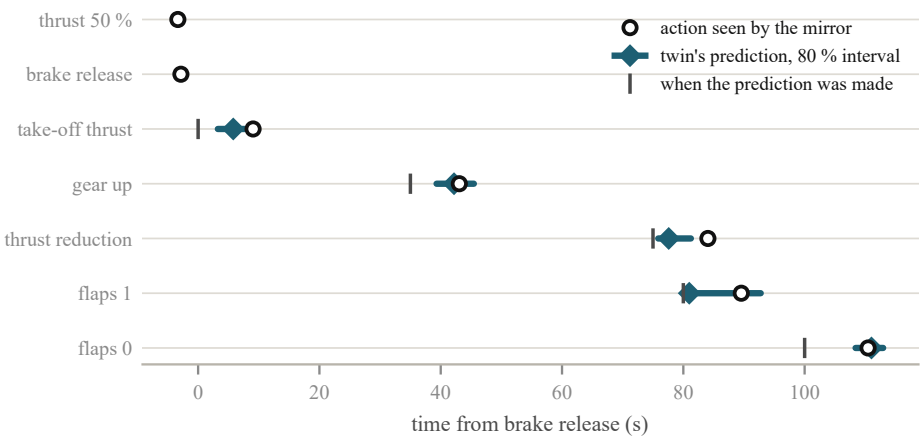


Fig. 1. One nominal take-off of validation pilot 11. Open circles: actions recognised by the mirror. Diamonds: what the twin predicted about ten seconds earlier, with its 80 % interval; the tick marks when the prediction was made. Four of the five actions fall inside their interval; the thrust reduction came 2.8 s after its interval closed. The first two actions precede the roll, when the twin does not yet predict.

on 8 November 2026. Corrections made after that date were judged again on the same pilots, no longer blind, and Table 2 marks those results as corrected.

Pre-registration. Before each computation, the criterion, its threshold and our predicted outcome were committed to the project's repository. When a criterion had to be amended, for instance because a rule of the evaluation was found to be wrong, the amendment was dated and the first judgement kept.

The pilot's own record, cross-validated. The comparisons of Section 5.4 use every pilot. To judge one pilot, the other eleven provide everything that is learned, and the pilot is predicted, flight after flight, from earlier flights alone. Where a prediction uses the pilot's record, it shrinks it toward the other pilots as in Eq. 1, but with a weight that is estimated rather than chosen: K is the ratio of within-pilot to between-pilot variance of the quantity, computed on the other eleven pilots by a one-way analysis of variance on values winsorised at their 5th and 95th percentiles [5]. Nothing in this evaluation is tuned.

Measures. A prediction is made at time t and judged on the pilot actions that come in the next h seconds, $h = 10$ or 20, leaving out actions that follow a failure the predictor could not yet have seen. All predictors are judged on the same pairs: 1,241 at 10 s and 1,938 at 20 s. The error is the median absolute difference between predicted and actual time, per pilot and pooled. We count the pilots for whom one predictor beats another, with a one-sided sign test, and give a 95 % interval for the pooled difference from 10,000 bootstrap resamples of pilots.

Table 2. Pre-registered criteria and their verdicts. *Frozen*: protocol frozen before any flight, pilots held out. *First run*: the first judgement of the criterion. *Corrected*: judged again after a correction, on data already seen. *Cross-val.*: all twelve pilots, each predicted without being learned from.

Criterion	Run	Result	Verdict
A pilot that learns: six modelled pilots against six held-out pilots, 60 flights			
Liftoff to gear up matches the pilots	frozen	11.6 s against 5.5 s, difference 6.1 s (99 % interval 4.8 to 7.8)	FAIL
Thrust reduction before the first flaps	frozen	57 of 57 flights against 34 of 35	PASS
Gear up before thrust reduction; omissions; slope of learning	frozen	too few events to decide (8 criteria)	inconclusive
Thrust reduction: height and time after liftoff	frozen	1,581 against 1,700 ft; 27.0 against 36.4 s	FAIL
The mirror, replaying all 211 take-offs			
Actions recognised (at least 90 %)	first run	1,680 of 1,686; 1,686 after correction	PASS
Median lag (at most 1.5 s)	first run	0.21 s; 0.08 s after correction	PASS
False recognitions (at most 10 %)	first run	4 of 1,684; 1 of 1,687 after correction	PASS
The twin			
Next action: the pilot's offset beats the generic model	first run	12 of 12 pilots	PASS
Next action: beats the other pilots' offset	first run	6 of 12 pilots	FAIL
same, with K set on calibration pilots	corrected	6 of 6 validation pilots	PASS
Twenty seconds ahead: beats reference B in error	first run	5 of 6 at 10 s and at 20 s	PASS
and predicts as many actions as B	first run	0.88 against 0.94 of actions at 10 s	FAIL
after a failure assumed in advance	first run	2 of 5 pilots	FAIL
an estimated cognitive state improves it	corrected	2 of 6 pilots; +0.012 s	FAIL
The twin against the pilot's own record			
Twin beats the pilot's own record	cross-val.	11 of 12 at 10 and 20 s	PASS
Twin beats the event baseline	cross-val.	9 of 12 at 20 s; 7 of 12 at 10 s	PASS (20 s)
the model alone, without the blend	cross-val.	10 of 12 at 20 s; 7 of 12 at 10 s	reported
Personalising the twin further improves it	cross-val.	5 of 12; worse by 0.07 s	FAIL
Flagging a missing action			
80 % intervals contain the action (70 to 90 %)	corrected	70 % of 983 actions	PASS
At most 0.75 false alerts per flight	corrected	0.09	PASS
Omitted action flagged within 10 s (80 %)	cross-val.	63 % (59 % without personalisation)	FAIL

5 Results

Table 2 lists every pre-registered criterion discussed in this paper with its verdict. We take the four questions in turn.

5.1 A pilot that learns, tested on pilots it never saw

The model learned. In development, one modelled pilot reproduced the shape of the human learning curve: the time from liftoff to gear up fell from 6.7 s on its first flight to 3.8 s on its second, once its compiled rules were reloaded, and then

stayed near 3.6 s; the pilots go from 15.1 to 5.7 and then 3.3 s. The frozen test, run on six modelled pilots of ten flights each against the six held-out pilots, did not confirm it. On the first flight the agent was close to the pilots, 16.1 s against 15.1 s, but it then stopped improving: 11.6 s on flights 2 to 7 and 11.5 s on flights 8 to 10, where the pilots reached 6.1 and 3.1 s. The pre-registered test of this delay fails. Its cause, among the documented differences between the development and frozen configurations, is not established. The order of the procedure held: the agent reduced thrust before the first flap retraction in all 57 flights where both occurred, as the pilots did in 34 of 35, and its rotation speed and peak pitch fell within the pilots' interquartile range. Its thrust reduction, on the other hand, came earlier and lower than the pilots', at 1,581 against 1,700 ft, a failed secondary criterion. Omissions were too rare on a nominal take-off to decide anything with six pilots on each side.

5.2 The mirror follows the pilots

We replayed all 211 take-offs into the live mirror, as if each were flown in front of it, and compared what it recognised with an independent offline reading of the same flights. On its first run, the mirror recognised 1,680 of the 1,686 actions found offline, with a median lag of 0.21 s. After one correction of how it holds an action, it recognised all 1,686, with a median lag of 0.08 s (at most 1.85 s), and reported a single action the offline reading did not see, a brake application during a rejected take-off, out of 1,687. Earlier, reading the nominal take-offs offline, the mirror had found that in 101 of 103 of them the pilots raised the gear before reducing thrust, the reverse of our written procedure; we corrected the procedure before the frozen test. A mirror that reads flights against the written procedure also checks the procedure against the flights.

5.3 Predicting the next action

When the mirror's model reaches a task, the pilot's own offset predicts the action better than the generic model for all twelve pilots, with a median error of 1.16 s against 3.76 s over 765 actions. Against the other pilots' offset it first won for only half of the pilots. Once the weight K of Eq. 1 was set on the calibration pilots, it won for each of the six validation pilots; over the 765 actions of all twelve pilots, the two errors were 1.16 and 1.45 s.

The look-ahead was first judged blind on 8 November 2026, on the held-out pilots. Its predicted times were closer than those of the reference B for five of six pilots at both horizons (1.64 against 2.24 s at 10 s, 1.85 against 2.57 s at 20 s). It failed, however, to predict as many of the pilots' actions as B did (0.88 against 0.94 of them at 10 s), it did no better than B after a failure assumed in advance, and it was too slow for live use. The last defect was corrected and judged again; the others remain. The version evaluated next is the one in service, which blends the model with the reference, feeds it the pilot's estimated cognitive state, and corrects the time base.

Table 3. Median error of each predictor, in seconds, on the same 1,241 and 1,938 prediction pairs (twelve pilots, each held out).

Predictor	10 s ahead	20 s ahead
The other pilots' median delay	2.46	2.77
The pilot's own record	1.97	2.48
Reference B (own record shrunk, $K = 16$)	2.03	2.42
Reference B, estimated weight	1.77	2.16
Event baseline: delay since the last event seen	1.53	1.80
Twin in service	**1.38**	**1.48**
Twin, personalised further	1.45	1.55

5.4 The twin against the pilot's own record

Table 3 compares the twin in service with five predictors that use no cognitive model. Four of them predict an action from a delay measured from an anchor: the median of the other pilots; the pilot's own median over earlier flights; the reference B used inside the twin; and B with the estimated weight of Section 4. The fifth, the *event baseline*, is the strongest we could build: it measures the delay from the last event already seen in the flight (liftoff, the failure, or the pilot's previous action), choosing among them the anchor whose delays vary least among the other pilots, and personalises that delay. It knows what has happened in the flight and to which procedure each action belongs, but not what the pilot has to do between two events.

The twin beat the pilot's own record for 11 of the 12 pilots at both horizons ($p = 0.003$), by 0.59 s at 10 s (95 % interval 0.18 to 0.99) and 1.00 s at 20 s (0.58 to 1.54). The comparison that matters more is with the event baseline. At 20 s the twin beat it for 9 of 12 pilots by 0.32 s (0.12 to 0.57): the count of pilots alone does not reach significance ($p = 0.07$), but the interval of the pooled difference excludes zero. At 10 s the twin won for only 7 of 12, and the interval, −0.11 to 0.33 s, includes zero. Most of the twin's advantage over the pilot's own record therefore comes from knowing what has happened in the flight: the event baseline alone beats the pilot's record for 11 of 12 pilots. The cognitive model adds a further 0.32 s when it looks twenty seconds ahead, where the next event is not yet known, and nothing we can establish at ten.

Since the twin in service blends the model with the reference B, the gain could come from the blend rather than from the model. We therefore compared the model alone, on the pairs for which it makes a prediction (1,213 at 10 s and 1,868 at 20 s). On its own, it beat the event baseline for 10 of 12 pilots at 20 s ($p = 0.02$), by 0.22 s (0.04 to 0.45), and for 7 of 12 at 10 s, where the interval again includes zero. The blend improves on the model alone for most pilots, but by less than 0.1 s, within the noise. The advantage at twenty seconds is the cognitive model's.

Figure 2a shows where the model matters most: on a pilot it does not yet know. On a pilot's first flight, with nothing learned about that pilot, the twin's error is

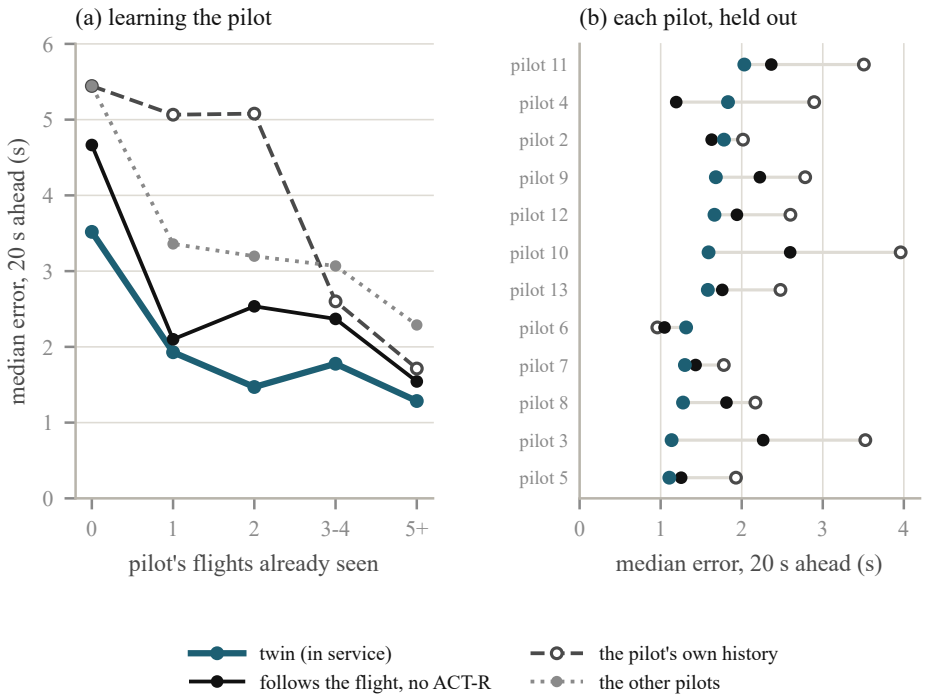


Fig. 2. (a) Median error 20 s ahead as a function of the number of the pilot's flights already seen, each pilot held out. (b) Each pilot's median error 20 s ahead; the twin is in colour.

3.5 s, against 4.7 s for the event baseline and 5.4 s for the other pilots' median; the model alone does better still, 3.0 s against 4.5 s for the event baseline on the pairs where both predict. The pilot's own record, which needs several flights to become useful, overtakes the other pilots' median only once three earlier flights are known. Personalising the twin itself further, with the estimated weight, made it slightly worse (1.55 against 1.48 s at 20 s), as the twin's offsets already draw on the pilot's record. Figure 2b gives the result pilot by pilot. The six validation pilots matter here, because their data never entered the twin's parameters: the twin beat the event baseline for five of them and the pilot's own record for all six.

5.5 What in a pilot's timing is personal

Why does the pilot's own record help for some actions and not others? The intraclass correlation of a delay, the share of its variance that lies between pilots rather than within one pilot's flights [18], answers directly (Fig. 3). The pacing of routine actions is a personal signature. How long a pilot waits between the 50 % thrust stabilisation and take-off thrust (0.71) and between brake release and take-off thrust (0.80) is stable from one flight to the next and differs between

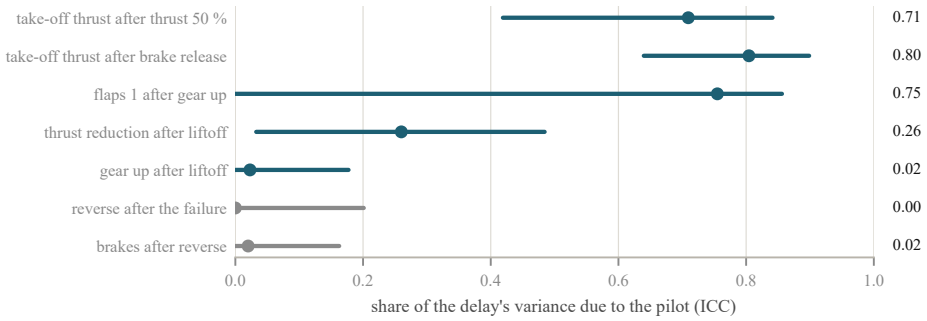


Fig. 3. Share of each delay's variance due to the pilot (intraclass correlation), twelve pilots, with 95 % intervals from a bootstrap over pilots. Routine actions in colour, responses to a failure in grey.

pilots, and so, though less certainly, is the delay between gear up and the first flap retraction (0.75, with a wide interval). The response to a failure is not: the delay between the failure and reverse thrust (0.00) and between reverse thrust and braking (0.02) is the same for everyone, within each pilot's own variation. Gear up after liftoff is not personal either (0.02): its delay varies as much from one flight to the next as from one pilot to another, in part because it shortens with practice (Section 5). A delay that changes with experience cannot be a fixed signature.

5.6 Flagging a missing action

A twin that predicts an action can flag it when it is overdue: an alert is raised once the time read exceeds the latest prediction made for the future by more than a margin learned on calibration pilots. On flights where the pilots made every action, the alerts are rare, 0.09 per flight. To test detection, we removed one key action at a time from what the mirror could see. Over the twelve pilots, 59 % of these simulated omissions were flagged within 10 s of the moment the action should have come, and 63 % when each pilot's delays were personalised, against the 80 % we had set. The split is clean. With personalised delays, omissions of take-off thrust, thrust reduction and reverse thrust were flagged in at least 96 % of cases, whereas gear up was flagged in 9 of 104 cases and the brakes of a rejected take-off in 8 of 54.

6 Discussion

What the cognitive model adds. A reader might expect a cognitive model to be needed to predict a pilot's next action, or might suspect that it adds nothing a careful statistic could not. Our results support neither view. Much of what makes the twin better than a pilot's own record is knowledge of the current flight, and

a baseline that has that knowledge comes close at a short horizon. The model's contribution is specific, and it holds for the model alone, without its blend with the reference: it predicts further ahead, beyond the next event, and it transfers to a pilot it has never seen, because the structure of the procedure, which it holds, is shared by all pilots while their timing is not. It also keeps every prediction inspectable: the twin can say which task it expects, which constraint it is waiting for, and from which earlier action it counts. A statistic that matches its error at ten seconds cannot say why.

Signature and response. That routine pacing is personal and failure response is not has two consequences. For training, it separates two kinds of measure: a tempo that is stable within a pilot, such as the pacing of the take-off roll, which characterises the pilot, and delays that shorten with practice, such as gear up after liftoff, which track progress. The delay from liftoff to gear up of the held-out pilots shrank from about 15 to 3 s over their first ten flights (Section 5); a twin that errs by 1.5 s twenty seconds ahead can follow a change of that size. For monitoring, an alert based on expectation can only be reliable for actions whose time is either personal and learned, or tightly cued by an event. The critical actions of our corpus, gear up and the brakes of a rejected take-off, are neither, which is why the alerts missed them. We also tried anchoring each alert on the event that triggers the action rather than on the clock; the anchor learned from one group of pilots did not transfer to another, and the attempt failed its criterion. Only the brakes, anchored on reverse thrust, were flagged reliably, and we have frozen that rule for the confirmatory study rather than adopt it after seeing the result.

What failed. Several criteria failed, and four of the failures carry a lesson. The learning model's skill stopped improving after its first flight in the frozen test, while it had not in development; a match that does not survive freezing is exactly what an evaluation of digital human models should catch. The look-ahead predicted fewer of the pilots' actions than the reference did, which the weighted average corrects only by construction. The cognitive state we estimate from physiology and behaviour changed the model's predictions but did not improve them: with the random seed of ACT-R held equal, the difference was $+0.012$ s (95 % interval -0.012 to $+0.036$). And the alerts fell short, for the reason given above.

Threats to validity. Twelve pilots are few. A majority of seven is weak evidence, and only a result such as 11 of 12 is strong on its own; we report both the count of pilots and an interval over pilots for that reason. The held-out pilots had been seen by the time several corrections were made, and the cross-validated comparisons of Section 5.4 were designed after an earlier attempt had failed; they are exploratory, however honest the estimation within them. The twin's own parameters were set on the calibration pilots, so for those six pilots the twin is not fully held out; restricted to the six validation pilots, the result is 5 of 6 against the event baseline. The mirror was checked against an offline reading of

the same flights, which applies the same procedural knowledge through a separate program, and, for take-off thrust, rotation and gear up, against an extraction made independently of our work; no flight was annotated by hand. Omissions were simulated by hiding an action that the pilot did make, so the rest of the flight is that of a pilot who did not forget. The pilots flew an A320 on a fixed-base simulator; only the learning test involves our agent's own flying, which is on an A330, and that test compares quantities that do not depend on the aircraft type. Finally, the data cannot be released, which limits replication to the protocol and the reported statistics.

Next. The decisive test is prospective. A study with new pilots, each flying twelve take-offs with failures, is planned; before any of its data exist, we have frozen the version of the twin it will test and three criteria: that the twin beats each pilot's own record, that it beats the event baseline at 20 s, and that the pacing of the routine, but not the response to a failure, is a personal signature.

7 Conclusion

The same declarative knowledge that lets an ACT-R pilot fly a procedure lets it read a pilot's flight against that procedure and anticipate the pilot's next actions. On twelve pilots, each predicted without being learned from, the twin anticipates a pilot's actions better than the pilot's own record, and better than the strongest statistical baseline we could build when it looks twenty seconds ahead. The pacing of routine actions is personal, the response to a failure is not. These results come with their failures, which we have kept in view: a model whose learning plateaued, alerts that miss the actions that matter most, and a cognitive state that does not help prediction. A confirmatory study with new pilots will tell which of these findings hold.

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